

Дурыгин Олег

Лекция 2. Гомография, или как визуализировать футбольные метрики на ТВ- трансляции

План

01

Лекция:
17:00 - 18:10
70 минут

02

QA сессия:
18:10 - 18:25
15 минут

03

Перерыв:
18:25 - 18:30
5 минут

04

Семинар:
18:30 - 19:15
45 минут



У тебя есть лекция?

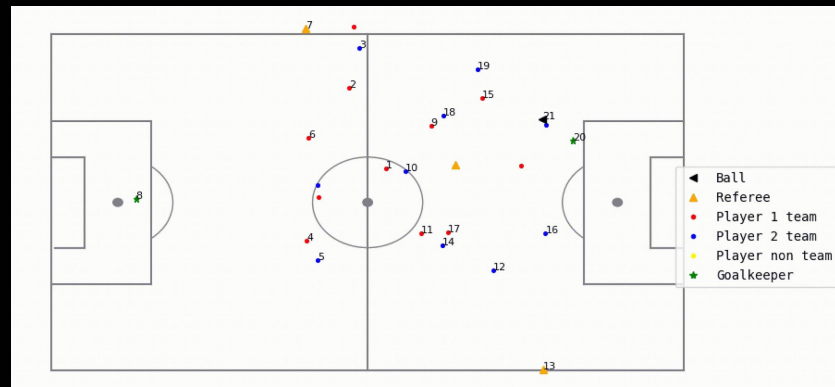
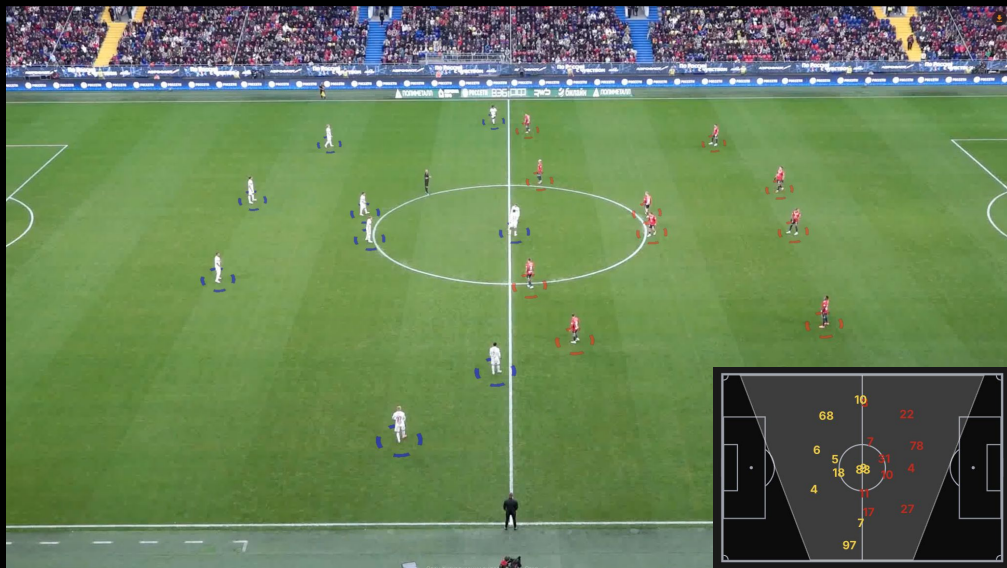
Лучше. У меня есть план лекции.

Введение + Ресар

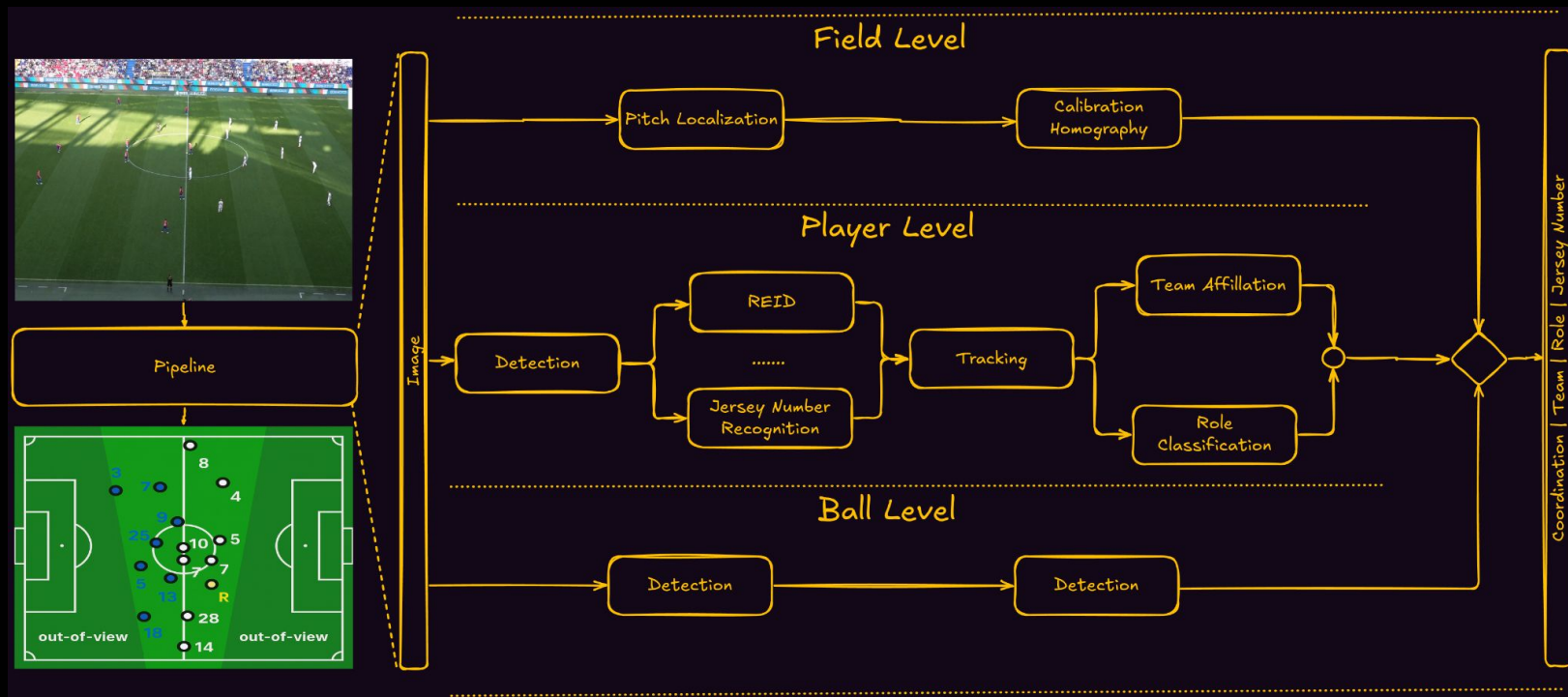
Типы данных в футболе



Трекинг данные

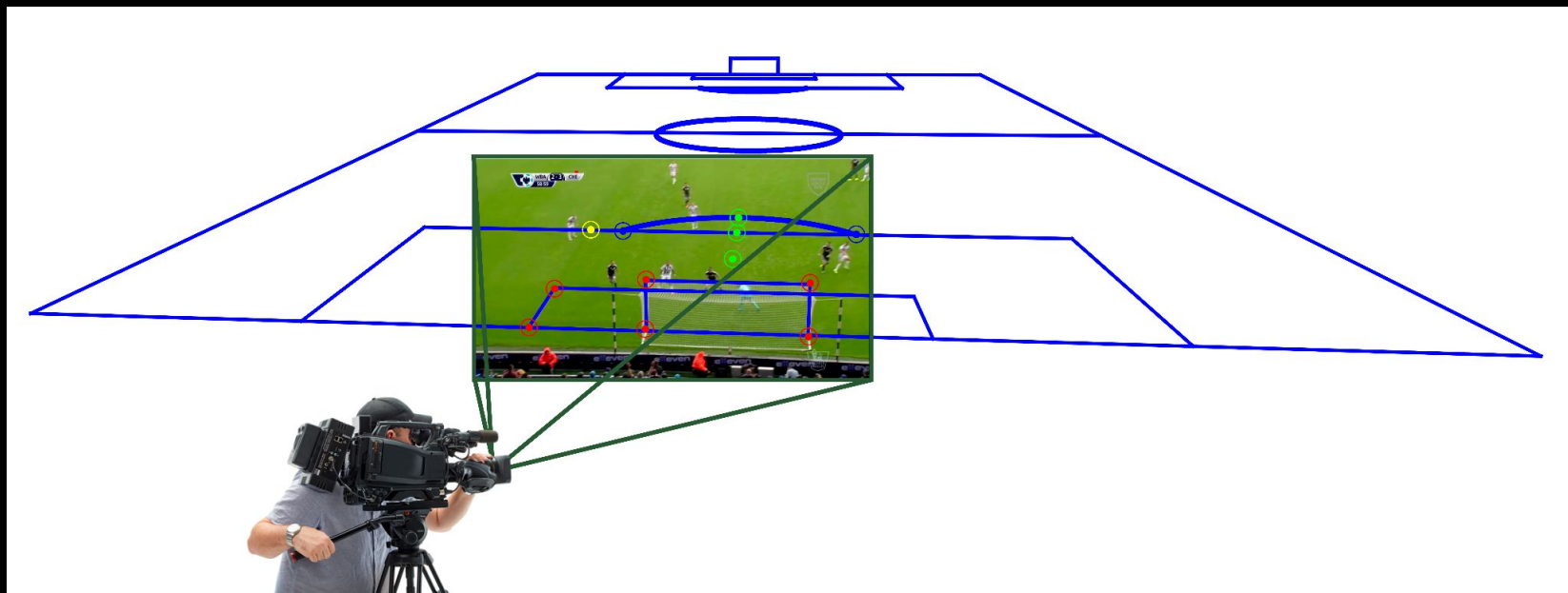


Постановка задачи

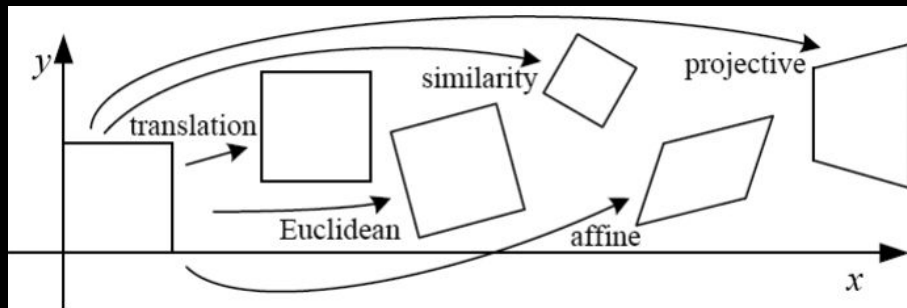


Гомография

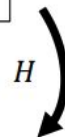
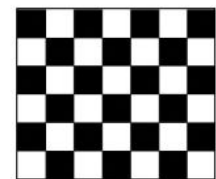
Гомография



Гомография



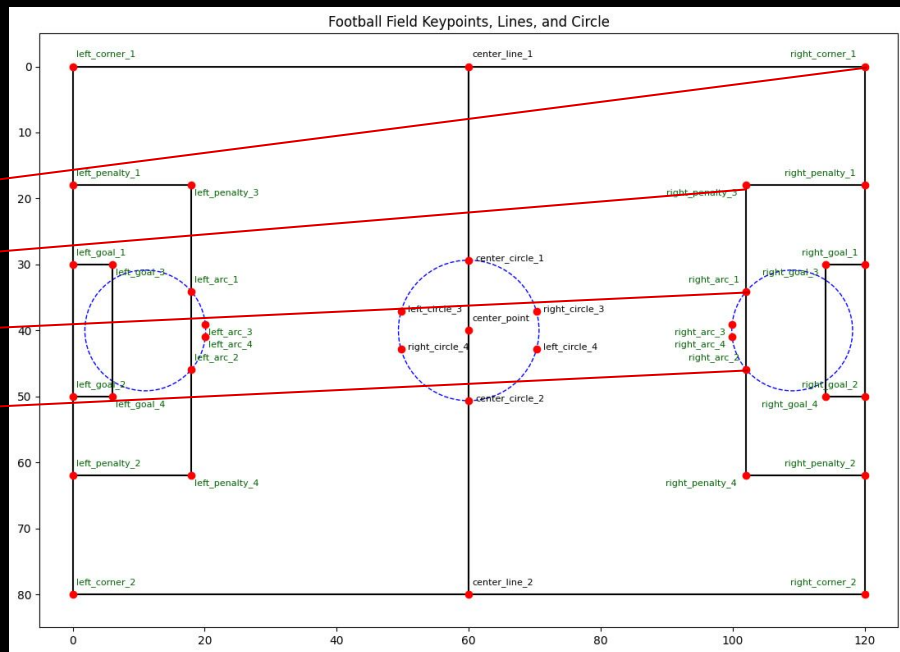
Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\begin{bmatrix} I & t \end{bmatrix}_{2 \times 3}$	2	orientation + ...	
rigid (Euclidean)	$\begin{bmatrix} R & t \end{bmatrix}_{2 \times 3}$	3	lengths + ...	
similarity	$\begin{bmatrix} sR & t \end{bmatrix}_{2 \times 3}$	4	angles + ...	
affine	$\begin{bmatrix} A \end{bmatrix}_{2 \times 3}$	6	parallelism + ...	
projective	$\begin{bmatrix} \tilde{H} \end{bmatrix}_{3 \times 3}$	8	straight lines	



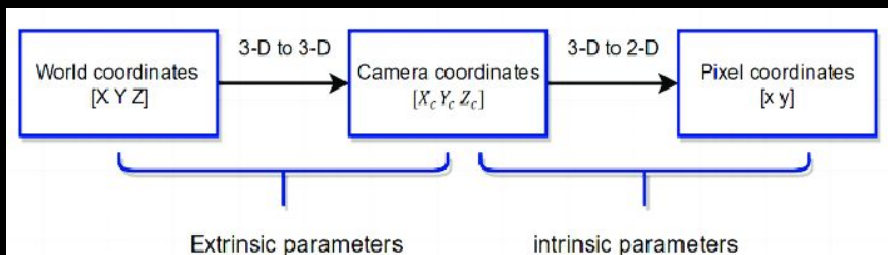
This matrix is called **Homography**

Гомография → cv2.findHomography

Минимум 4 пары точек

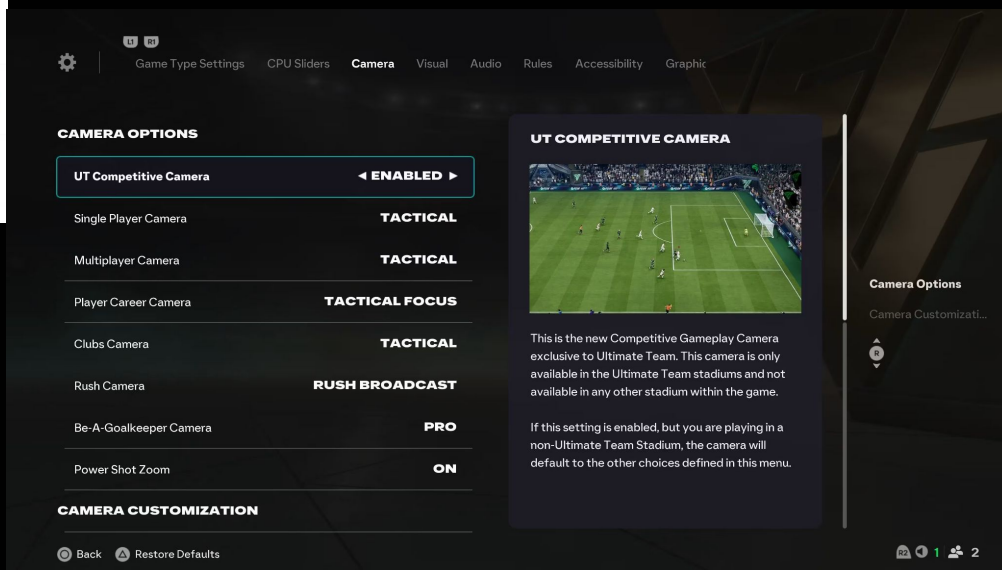


Параметры камеры

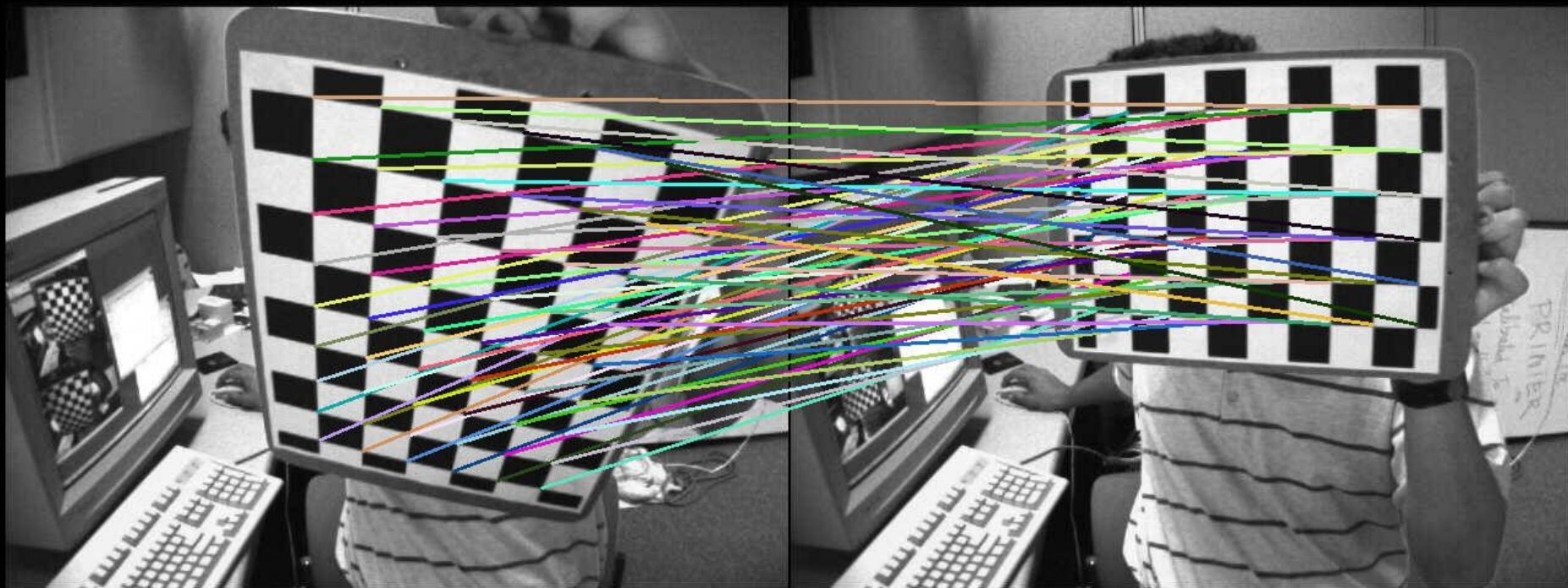


$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

2D Image Coordinates Intrinsic properties (Optical Centre, scaling) Extrinsic properties (Camera Rotation and translation) 3D World Coordinates



Калибровка → `cv2.calibrateCamera`



Дисторсия камеры → cv2.undistort

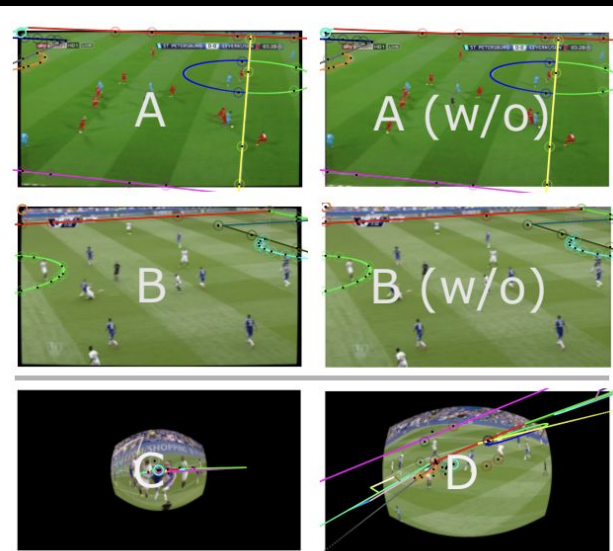
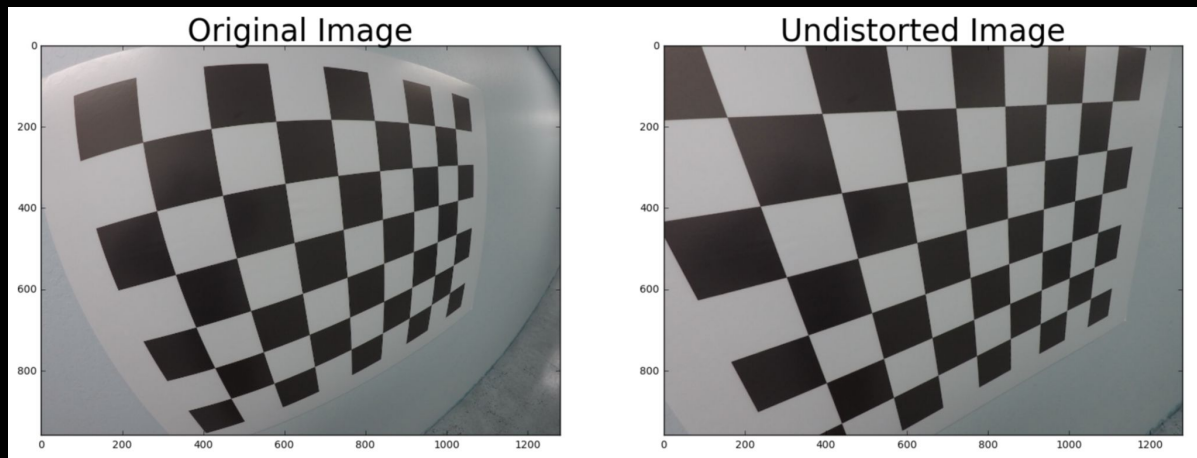
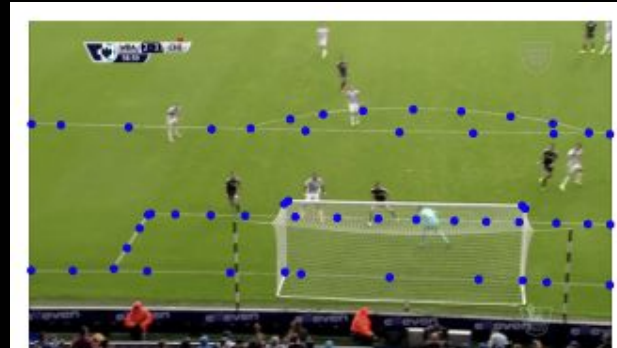
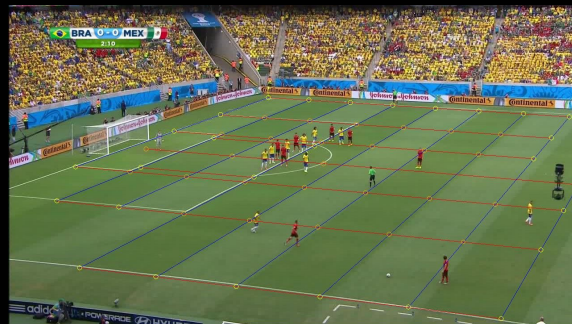
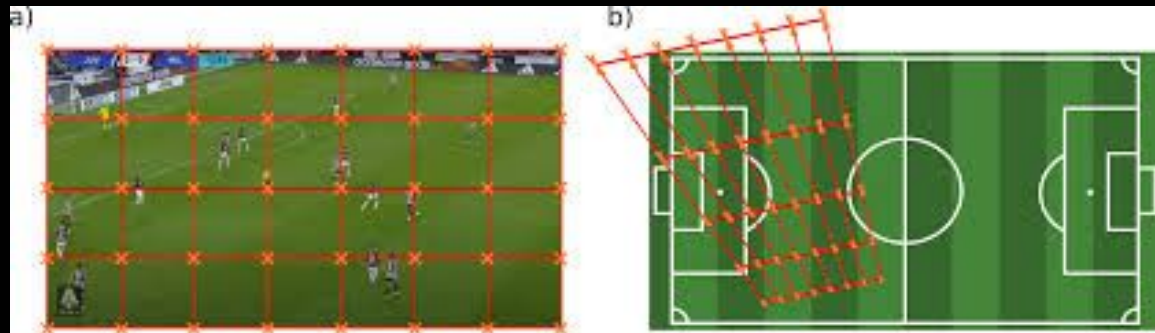
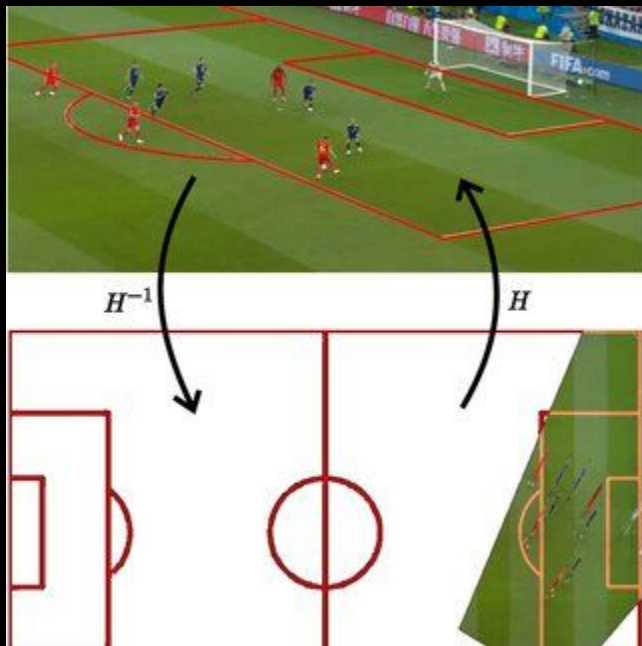


Figure 5: A, B: Samples where radial lens distortion is present (left: with correction, right: without correction for comparison); C, D: samples with lower *FoV* result in trivial local minima when jointly optimizing distortion coefficients and camera parameters.

Хрупкость гомографии



Метрики

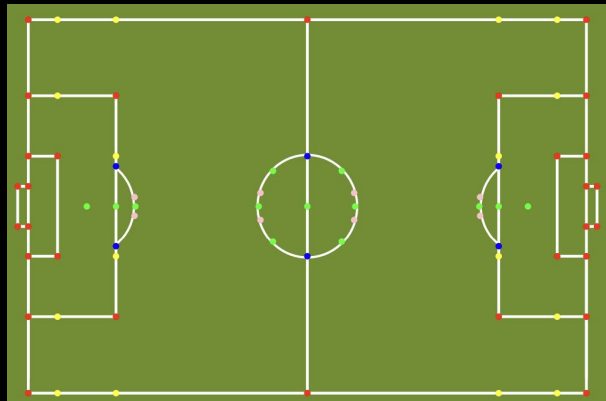
Метрика Jaccard

Jaccard @ 5 - сравнение gt точек в плоскости кадра с точками полученными после репроекции макета поля с помощью гомографии. True Positive считается - min euclidean distance от gt точки до pred line с threshold 5.

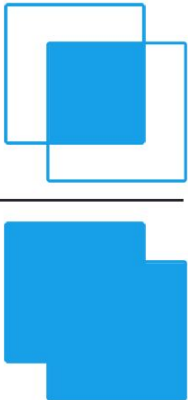
completeness - количество кадров, на которых мы смогли оценить гомографию или параметры камеры

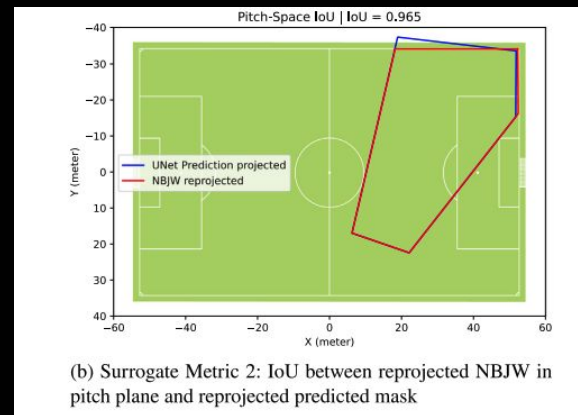
final score - completeness * Jaccard @ 5

$$\text{JaC}_{\gamma} = \frac{\text{TP}_{\gamma}}{\text{TP}_{\gamma} + \text{FN} + \text{FP}},$$



Метрика IoU

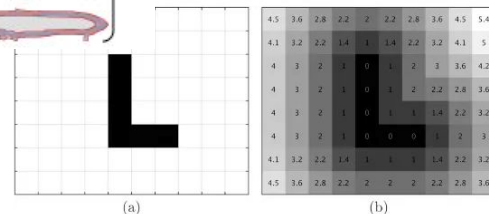
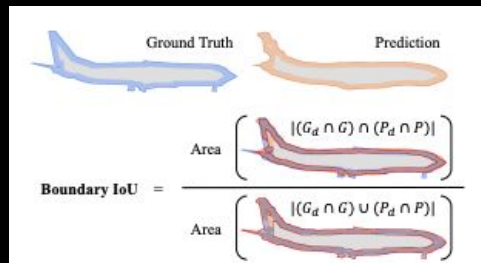
$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$




Экспериментальные метрики

Boundary IoU - оценивает глобальное перекрытие буферизованных линий, но почти не реагирует на локальные сдвиги границ, особенно при длинных и тонких объектах.

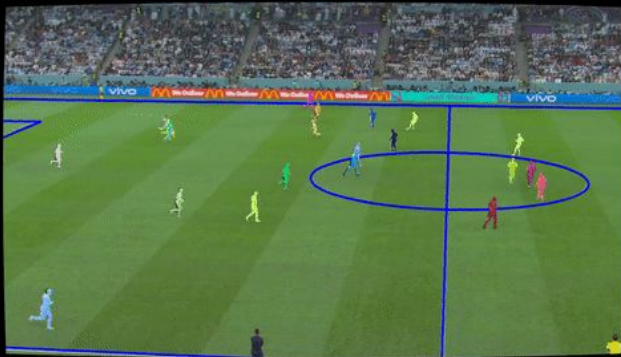
Chamfer distance - измеряет среднее расстояние между точками эталона и предсказания, отлично отражая локальные ошибки и пропуски.



In computer vision and 3D graphics, Chamfer distance is a distance function that estimates the similarity between two point clouds. For two point clouds $P = \{p_1, p_2, \dots, p_m\}$ and $Q = \{q_1, q_2, \dots, q_n\}$, the chamfer distance is defined as follows:

$$\text{Chamfer Distance}(P, Q) = \sum_{p \in P} \min_{q \in Q} \|p - q\|_2^2 + \sum_{q \in Q} \min_{p \in P} \|q - p\|_2^2$$

Экспериментальные метрики



Evaluation Criteria

Submissions will be evaluated using the following metrics:

- Global MPJPE (Mean Per Joint Position Error): Measures the absolute error per joint between the prediction and ground truth.
- Local MPJPE (Mean Per Joint Position Error relative to root joints): Captures local accuracy by evaluating joint positions relative to the root joint. Unlike PA-MPJPE, no scaling or rotation is applied.

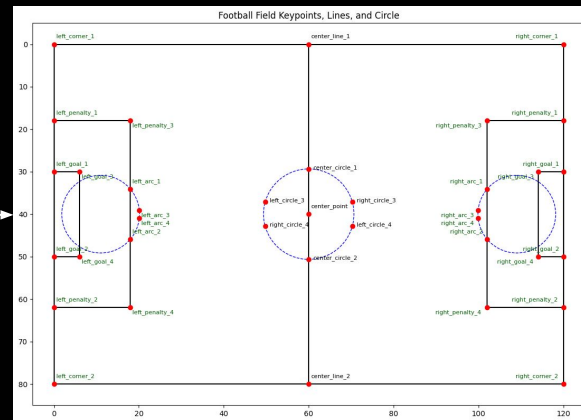
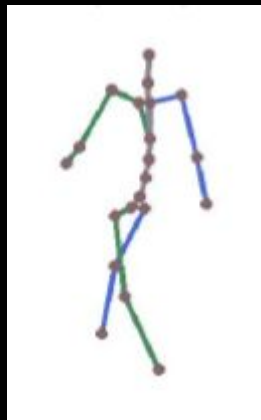
The final score is computed as:

$$\text{Final Score} = \text{global MPJPE} + 5 \times \text{Local MPJPE}$$

The weighting factor 5 ensures that both terms contribute meaningfully. As local MPJPE is typically an order of magnitude smaller, this scaling brings them to a comparable range.

Экспериментальные метрики

производная)



$$MPJPE = \frac{1}{N_F} \frac{1}{N_J} \sum_{f,j} \|p_{f,j} - \hat{p}_{f,j}\|_2$$

$$v_{j,t} = P_{j,t+1} - P_{j,t}$$

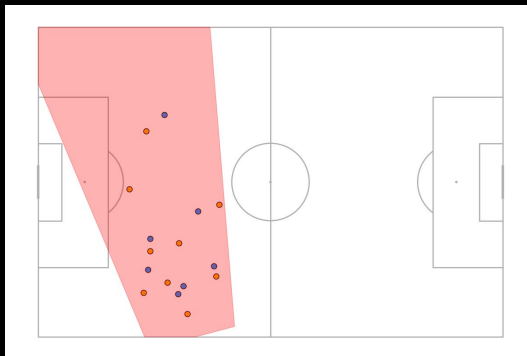
$$MPJVE = \frac{1}{T-1} * \frac{1}{J} \sum_{t=1}^{T-1} \sum_{j=1}^J \|v_{t,j} - \bar{v}_{t,j}\|_2$$

$$a_{j,t+1} = P_{j,t+2} - 2 * P_{j,t+1} + P_{j,t}$$

$$MPJAccE = \frac{1}{T-2} * \frac{1}{J} \sum_{t=1}^{T-2} \sum_{j=1}^J \|a_{t,j} - \bar{a}_{t,j}\|_2$$

Существующие решения

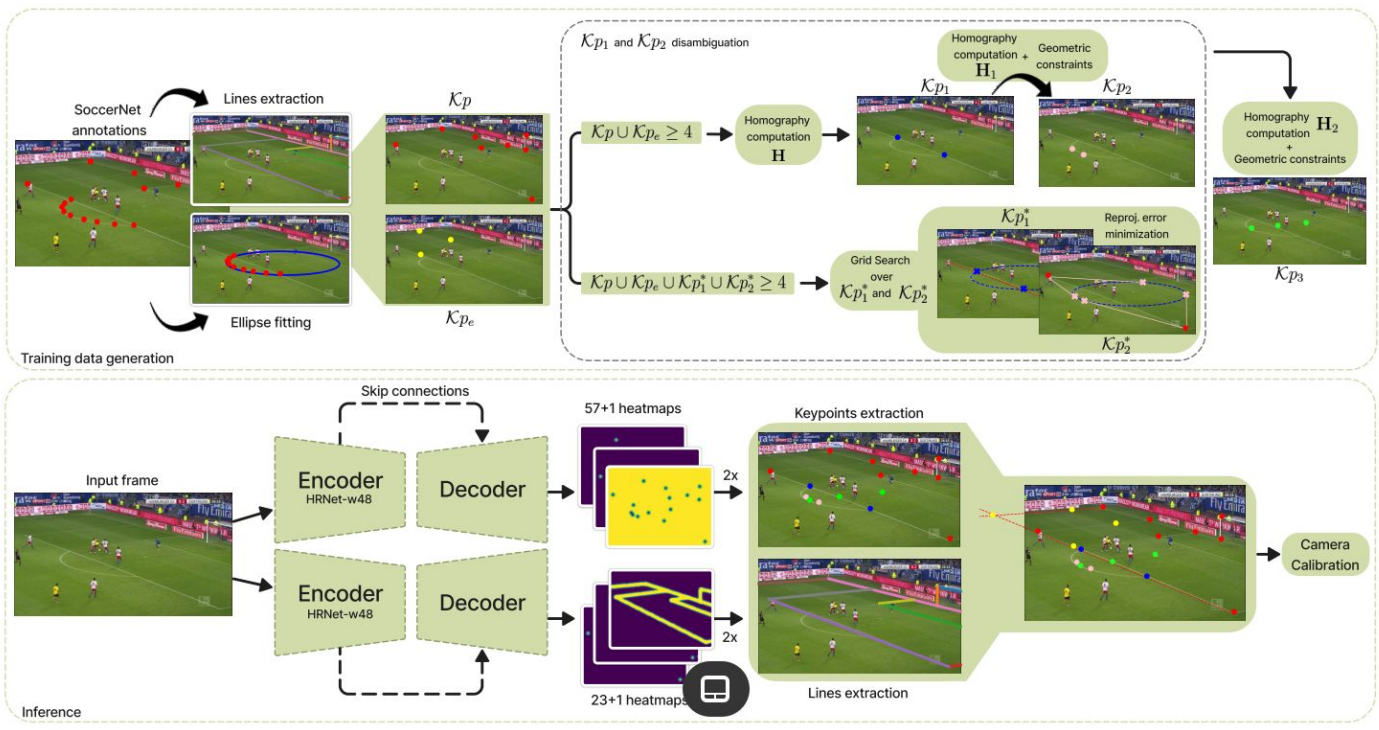
Сложность задачи CV с одной камерой



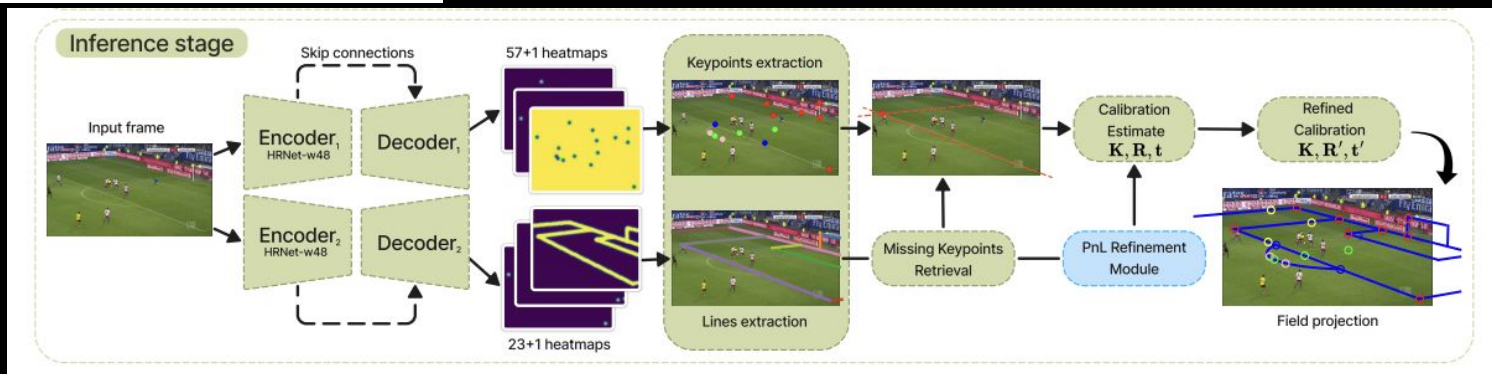
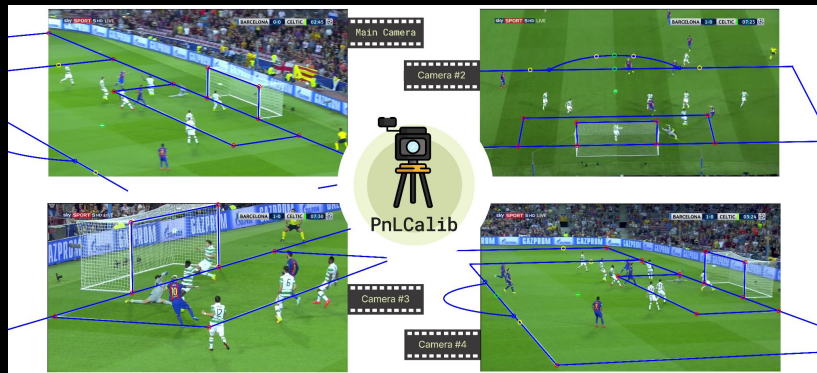
Сложность задачи CV с одной камерой



nbjw calib



PnLcalib



BroadTrack
Broadcast Camera Tracking for Soccer

The diagram illustrates the BroadTrack architecture, which processes two consecutive frames, I_{t-1} and I_t , to track a soccer ball.

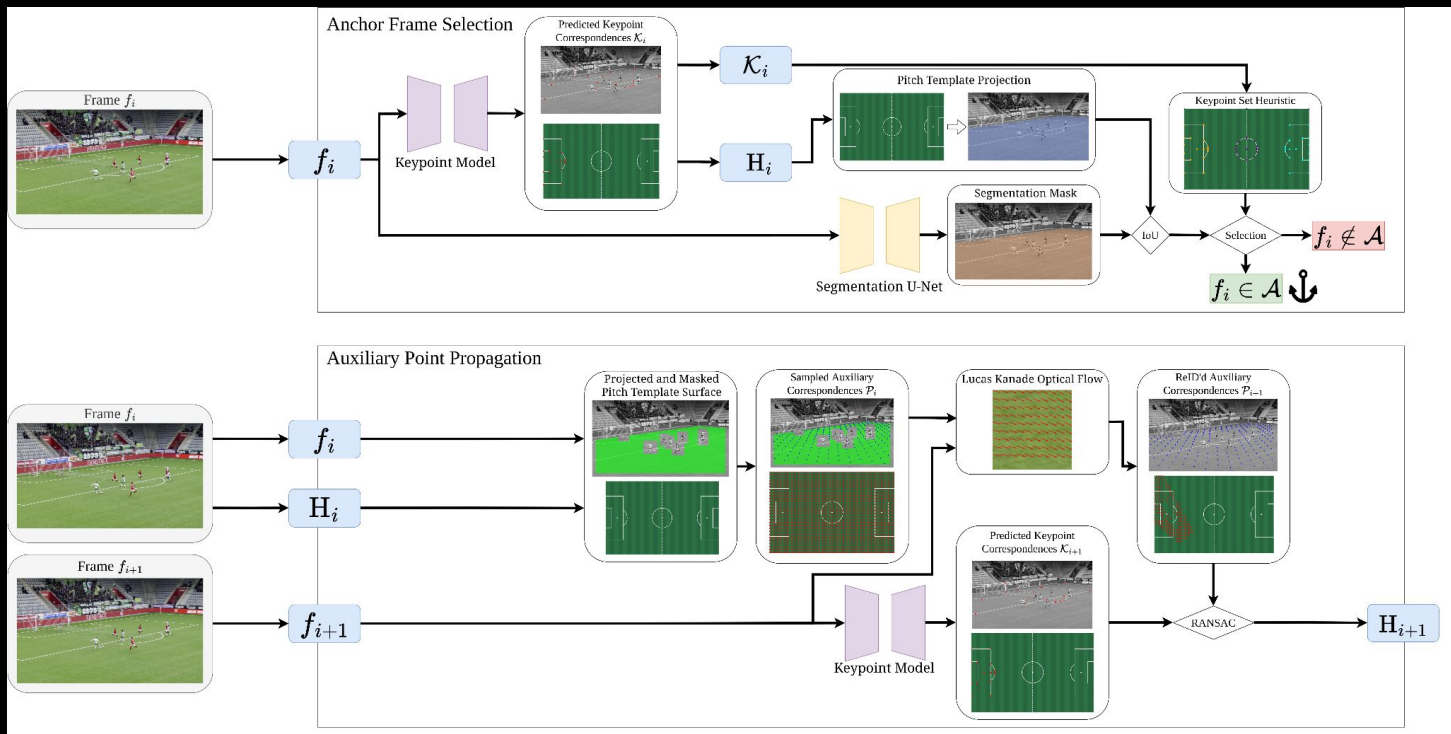
Input and Initial Processing: The input frames I_{t-1} and I_t are processed by two parallel modules: **Filtered optical flow** and **Semantic line detector**. The optical flow module outputs a visualization of yellow motion vectors, and the semantic line detector outputs a visualization of red and green lines representing the field boundaries.

Update Module: The outputs from the previous modules are fed into the **Update** module. This module contains two sub-processes: **Point reprojection error** and **Line reprojection error**. These errors are used to refine the camera pose. The **Update** module also receives the previous camera pose κ_{t-1} as input. The output of the **Update** module is the current camera pose κ_t .

Reinitialization Module: The **Update** module also feeds into the **Reinitialization** module. This module contains a **Keypoint detector** and a **2-point estimation** block. The **Keypoint detector** identifies key points in the current frame, which are then used by the **2-point estimation** block to estimate a new camera pose. The **Reinitialization** module outputs a new camera pose κ_t when the condition $s_t < \tau$ is met.

Output: The final output of the system is the current camera pose κ_t , which is used to track the soccer ball in the current frame I_t .

AuxFlow



Применение

Применение графики в ЛаЛиге



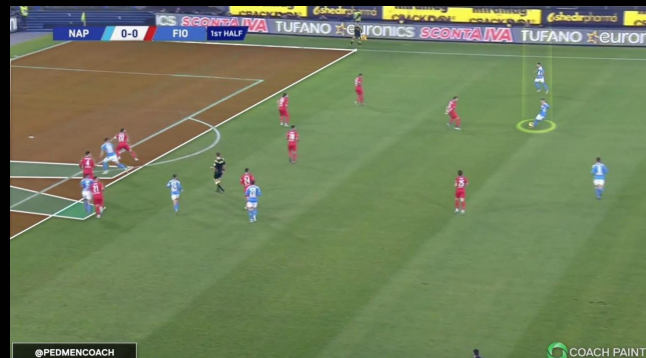
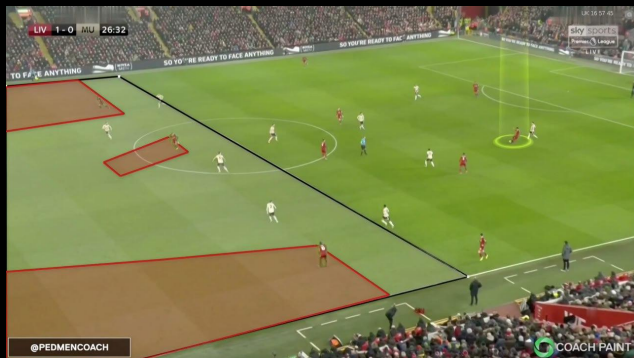
Применение графики в Бундеслиге



Применение графики в Бундеслиге



Coach Paint



Всем спасибо за внимание!

